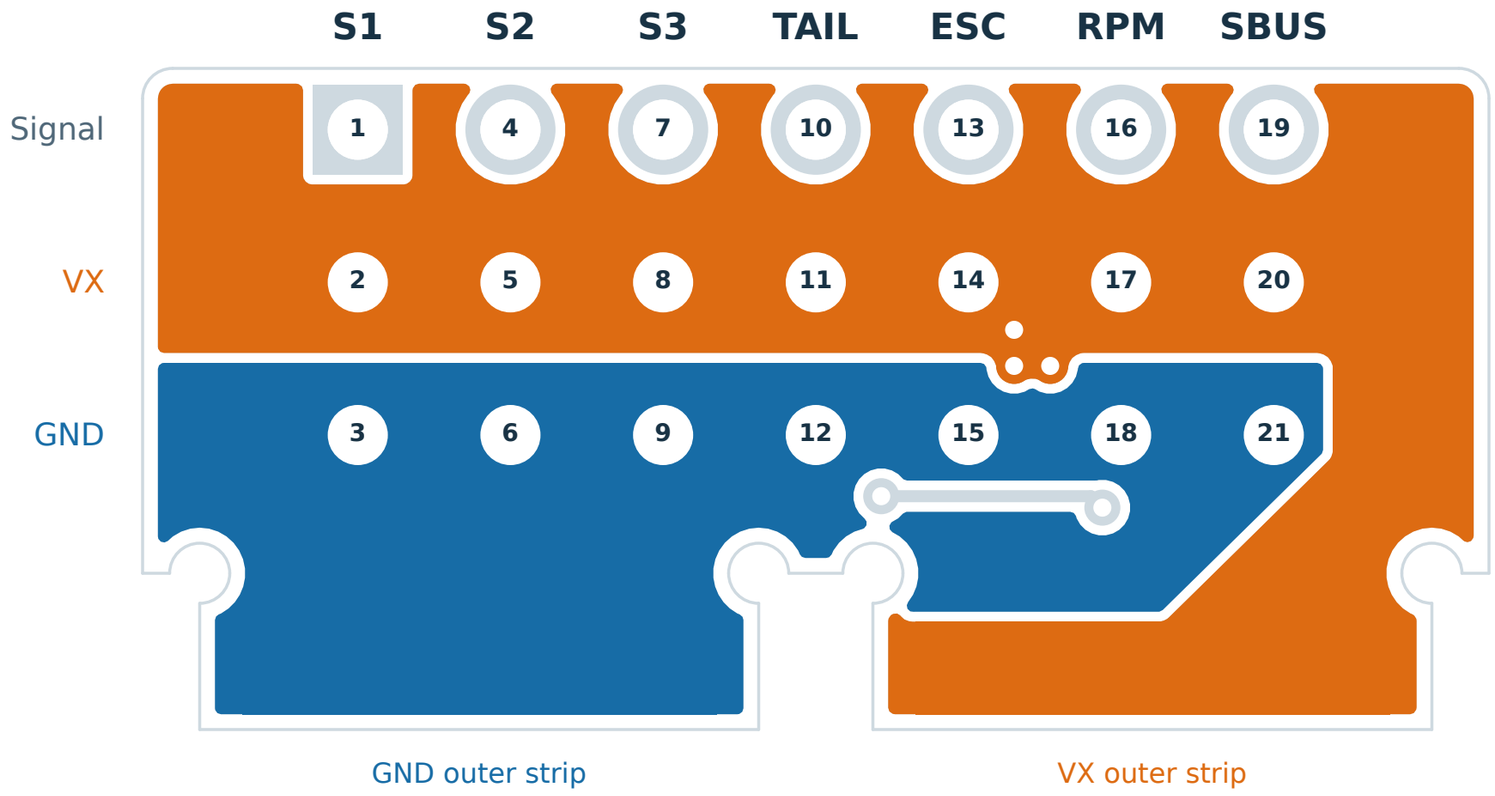
AFTER / compact two-layer power + matched-fit prototype

One continuous VX front sheet. Standard servo interface unchanged. Engineering prototype, not a current or flight release.

F.Cu | VX orange, GND blue, signals gray

Exact saved fill: one connected VX polygon and one connected GND polygon.



What changed

Broad VX copper bypasses the old four-servo daisy-chain and now joins the ESC, RPM and SBUS columns.

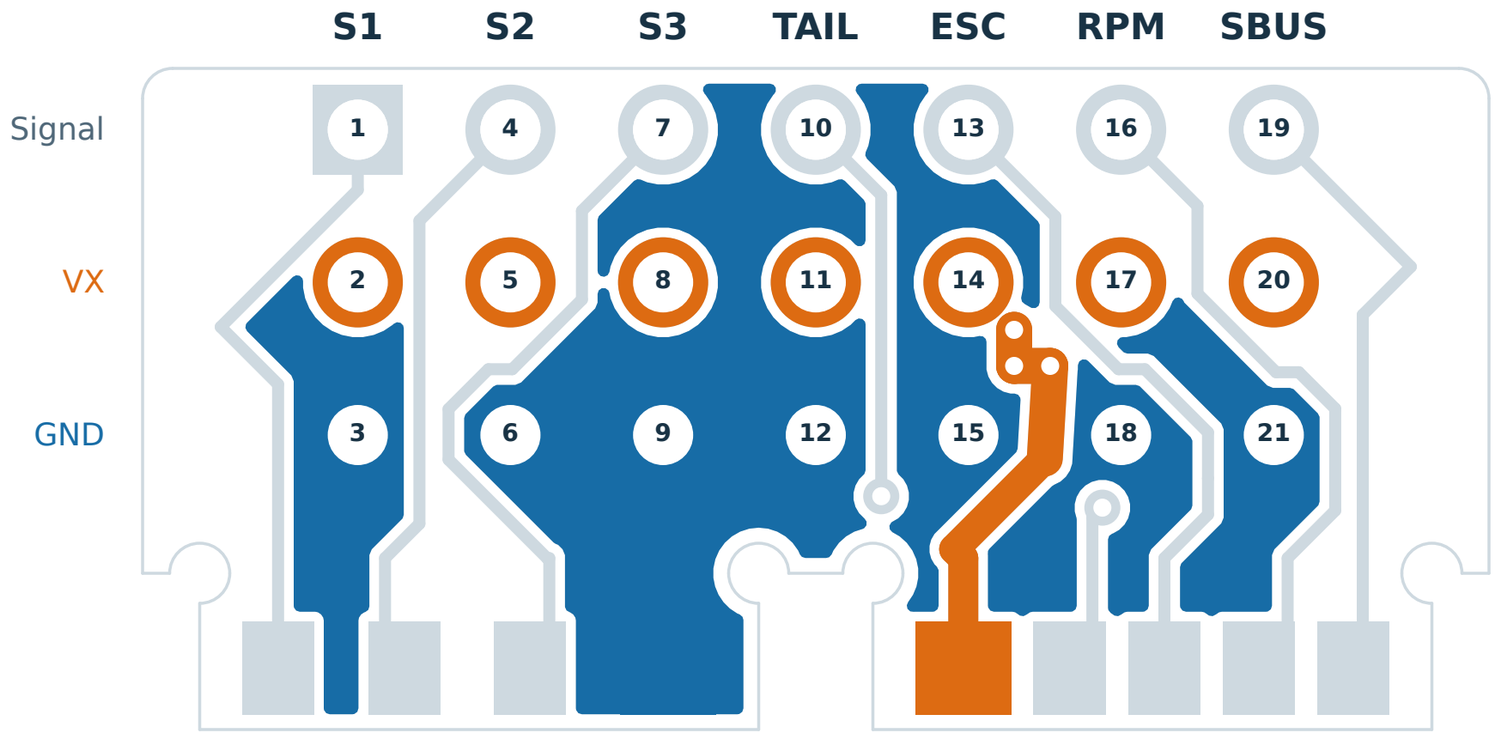
ESC signal moved to B.Cu.
The old shared 0.50 mm VX back-layer V-bridge is removed.

Three ordinary plated VX vias join the main-board branch to the sheet.
No filled or capped vias assumed.

Nominal 2 oz copper on both faces.
60 µm minimum is a proposed acceptance requirement, not a guaranteed public process tolerance.

B.Cu | return geometry and main-board branch

Back viewed from above, not mirrored. Five GND fills are connected through actual copper/PTH paths.



Tabs: 9.3 mm wide × 2.6 mm long; 270° R0.5 root reliefs. Mating slots: 9.8 × 2.10 mm R0.5.

What is still conditional

Copper modeling can apply 10 A at the pad ports. The retained header lists 3 A; factory female contacts and BEC leads are unknown.

Both feeds share the same BEC.
Single-feed-loss 10 A operation is not a qualified system rating.

Close fit requires actual finished profile/thickness inspection and matched-pair gauging. No external brace or new harness is selected.

Native child DRC: 0 errors / 0 opens.
22 inherited advisories remain visible.

22.4 × 11.0 mm child envelope including tabs • 2.54 mm header pitch • 1.50 mm pads / 1.00 mm holes

New VX vias: 0.60 mm pad / 0.30 mm drill • Minimum spacing 0.16 mm • Copper-to-edge 0.254 mm

Source PCB SHA-256: b49ef4108ab0b85d4cd96fac22553e3bc16b214f5acde148a8680e1212097ee6

KiCad 10.0.7 native saved-fill plots; colors overlaid from exact saved polygons, tracks and pads. No physical measurement implied.